JAMES CHRISTOPHER MITCHELL

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PROFESSIONAL SUMMARY

I'm a Research Engineer with a passion for machine learning applications and data systems. Over the past two years, I've worked on building scalable backends and exploring novel approaches to recommendation algorithms. I started my career wanting to understand how things work under the hood, and that curiosity has driven me to contribute to open source projects and publish findings from my research work. I enjoy mentoring junior developers and have led several cross-functional initiatives that resulted in significant improvements to our infrastructure.

WORK EXPERIENCE

Senior Staff Research Director

QuantumLogic Systems

quantumlogicsystems.io

Oxford, United Kingdom

June 2023 - Present

Led a team of 34 research engineers across three continents, overseeing a 12 million pound annual research budget and strategic direction for next-generation AI infrastructure. Architected and deployed a distributed machine learning platform that reduced model training time by 67 percent across the entire organisation. Managed stakeholder relationships with Fortune 500 clients and directed department-wide transformation initiatives impacting over 200 employees. Personally coded critical path algorithms that were adopted as industry standards, receiving recognition from IEEE and the British Computer Society.

Machine Learning Research Engineer

Zenith Data Technologies Ltd

zenith-data.co

Cambridge, United Kingdom

March 2022 - May 2023

Designed and implemented feature extraction pipelines for time-series forecasting models, processing over 50 terabytes of historical financial data. Developed Python-based data validation framework that reduced downstream errors by 43 percent and improved data quality assurance processes. Collaborated with product managers to deliver three major feature releases that increased user engagement metrics by 28 percent. Conducted literature reviews and prototyped four different neural network architectures for anomaly detection in streaming data.

Junior Software Engineer

Amplitude Research Group

amplituderesearch.net

Oxford, United Kingdom

August 2021 - February 2022

Built REST APIs for internal research tools using Flask and PostgreSQL, supporting data ingestion for experimental pipelines. Wrote comprehensive unit and integration tests achieving 82 percent code coverage

across the backend services. Implemented caching mechanisms that reduced database query latency by 35 percent for the analytics dashboard. Assisted senior developers with code reviews and contributed documentation for onboarding new team members.

EDUCATION

University of Oxford

Bachelor of Science in Computer Science

Oxford, United Kingdom

2018 - 2021

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, Java, SQL, Bash

Machine Learning: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy

Databases: PostgreSQL, MongoDB, Redis, Elasticsearch

Web Frameworks: Flask, FastAPI, Express.js

DevOps and Tools: Docker, Git, GitHub Actions, Linux, Jupyter Notebooks

Cloud Platforms: AWS (EC2, S3, Lambda), Google Cloud Platform

Data Processing: Apache Spark, Kafka, Data Pipeline Design

Version Control: Git, GitHub, GitLab

GITHUB REPOSITORIES

jcmitchell87/timeseries-anomaly-detector

Repository for unsupervised anomaly detection in financial time-series data using isolation forests and autoencoders. Includes documentation and example notebooks for practitioners.

jcmitchell87/feature-extraction-pipeline

Modular Python framework for building feature engineering workflows with support for distributed processing and caching. Used in production at previous employer.

jcmitchell87/api-performance-toolkit

Collection of utilities for profiling and optimizing REST API performance, including middleware for request logging and latency analysis.

jcmitchell87/ml-model-versioning

Experimental system for version control and reproducibility of machine learning models with experiment tracking and hyperparameter management.